

RATE & EXTENT – CALCULATING RATES

KEY VOCABULARY

Mean rate: amount of reactant used or product made ÷ time.

Tangent: a straight line touching a curve to find the rate.

Gradient: change in y ÷ change in x of a tangent.

Units: g/s, cm³/s or mol/s depending on what is measured.

COMMON MISCONCEPTIONS

~~Rate is constant throughout.~~

✓ Rate is fastest at the start and slows down.

~~A steeper curve means a slower rate.~~

✓ A steeper gradient means a faster rate.

~~Rate is read straight off a curve.~~

✓ Draw a tangent and find its gradient.

MEAN RATE

$$\text{mean rate} = \frac{\text{quantity of reactant or product}}{\text{time taken}}$$

CONVERSION REMINDER

Rate is fastest at the start

1 MEAN RATE (MASS)

A flask loses 6.0 g in 30 s. **Calculate** the mean rate of reaction. [2 marks]

Answer: _____ g/s

2 MEAN RATE (GAS)

60 cm³ of gas is made in 40 s. **Calculate** the mean rate. [2 marks]

Answer: _____ cm³/s

3 USING A TANGENT

Describe how to find the rate at a given time from a curved graph. [2 marks]

4 WHY IT SLOWS

Explain why the rate decreases as the reaction continues. [2 marks]

5 FILL THE GAPS

Complete the sentence using the word bank. [2 marks]

Draw a _____ to the curve and find its _____ to get the rate.

gradient

start

tangent

end

6 READ A GRAPH

A graph levels off at 80 cm³. **State** what this shows about the reaction. [1 mark]

